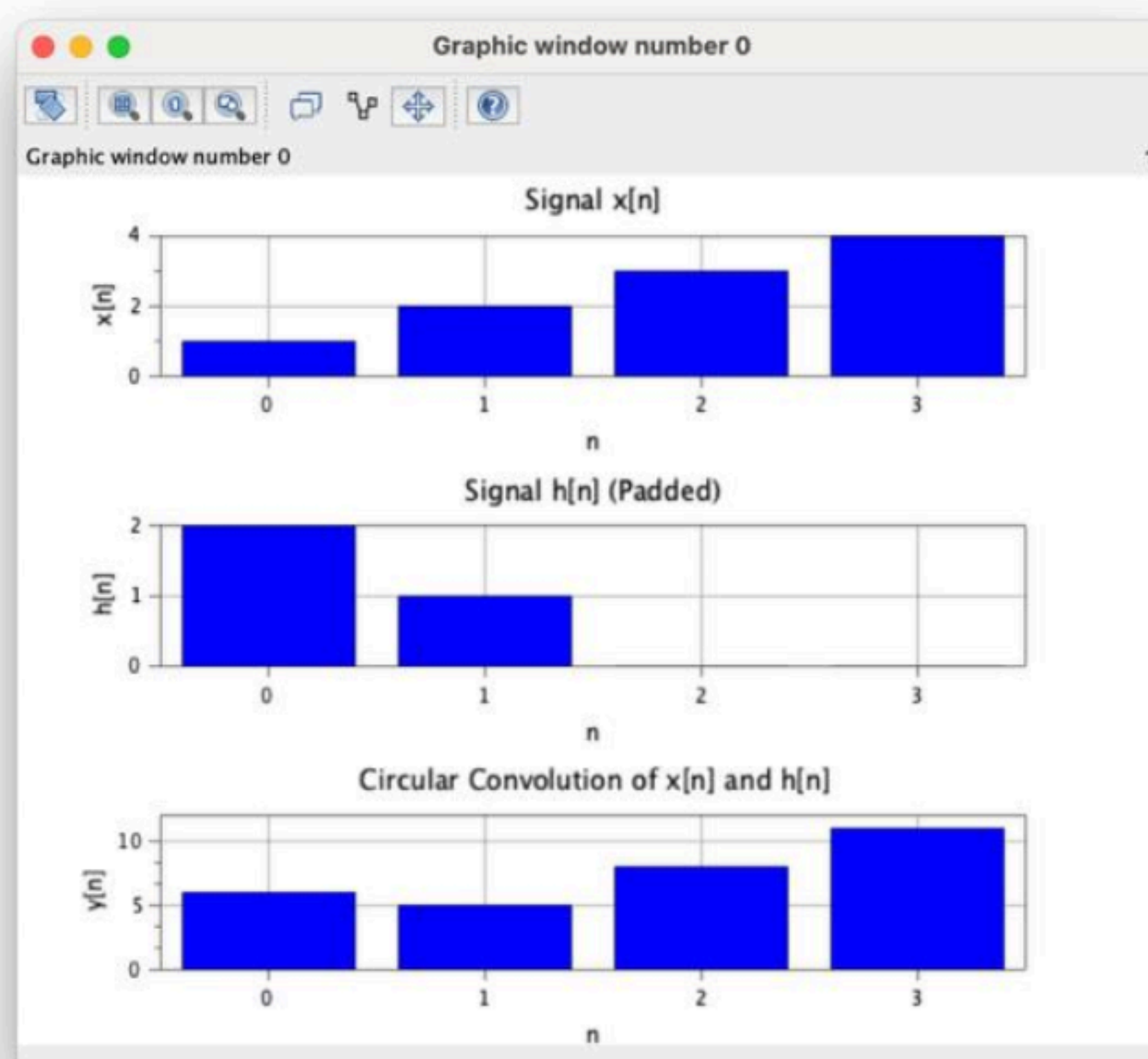


signals4.1.sce x signals 4.2.sce x signals 5.1.sce x signals 5.2.sce x signals 6.1.sce x 6.2.sce x signals 7.1.sci x 7.2.sce x 8.1.sce x 8.2.sci x 9.1.sce x

```

1 clc;
2 clear;
3 clf();
4
5 // Signals
6 x = [1 2 3 4];
7 h = [2 1 0 0];
8
9 // Circular Convolution
10 X = fft(x,-1);
11 H = fft(h,-1);
12 Y = X .* H;
13 y = real(ifft(Y));
14
15 n = 0:3;
16
17 // Plot x[n]
18 subplot(3,1,1);
19 bar(n,x);
20 xtitle("Signal-x[n]","n","x[n]");
21 xgrid();
22
23 // Plot h[n]
24 subplot(3,1,2);
25 bar(n,h);
26 xtitle("Signal-h[n]-(Padded)","n","h[n]");
27 xgrid();
28
29 // Plot Circular Convolution
30 subplot(3,1,3);
31 bar(n,y);
32 xtitle("Circular-Convolution-of-x[n]-and-h[n]","n","y[n]");
33 xgrid();
34

```



signals 5.2.sce x signals 6.1.sce x 6.2.sce x signals 7.1.sci x 7.2.sce x 8.1.sce x 8.2.sci x 9.1.sce x 9.2.sce x signals1.2.sce x

```
1 clc;
2 clear;
3 // Input sequence (length should be power of 2)
4 x = [1, 2, 3, 4, 0, 0, 0, 0];
5 N = length(x);
6 X = fft(x, -1); // Scilab built-in FFT (DIT equivalent)
7 // Display results
8 disp("FFT (DIT) Result:");
9 disp(X);
10 subplot(2,1,1);
11 plot(0:N-1, abs(X), 'r');
12 xlabel('Frequency Index');
13 ylabel('Magnitude');
14 title('Magnitude Spectrum (DIT FFT)');
15 subplot(2,1,2);
16 phaseX = atan(imag(X), real(X)); // Phase in radians
17 plot(0:N-1, phaseX, 'b');
18 xlabel('Frequency Index');
19 ylabel('Phase (radians)');
20 title('Phase Spectrum (DIT FFT)');
21
```

